

An Introduction to 3D Printing

Part 2: Practical Application

Examples, Critical Thinking, and Design

Outline

Part 1: Printers

Process

Limitations

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

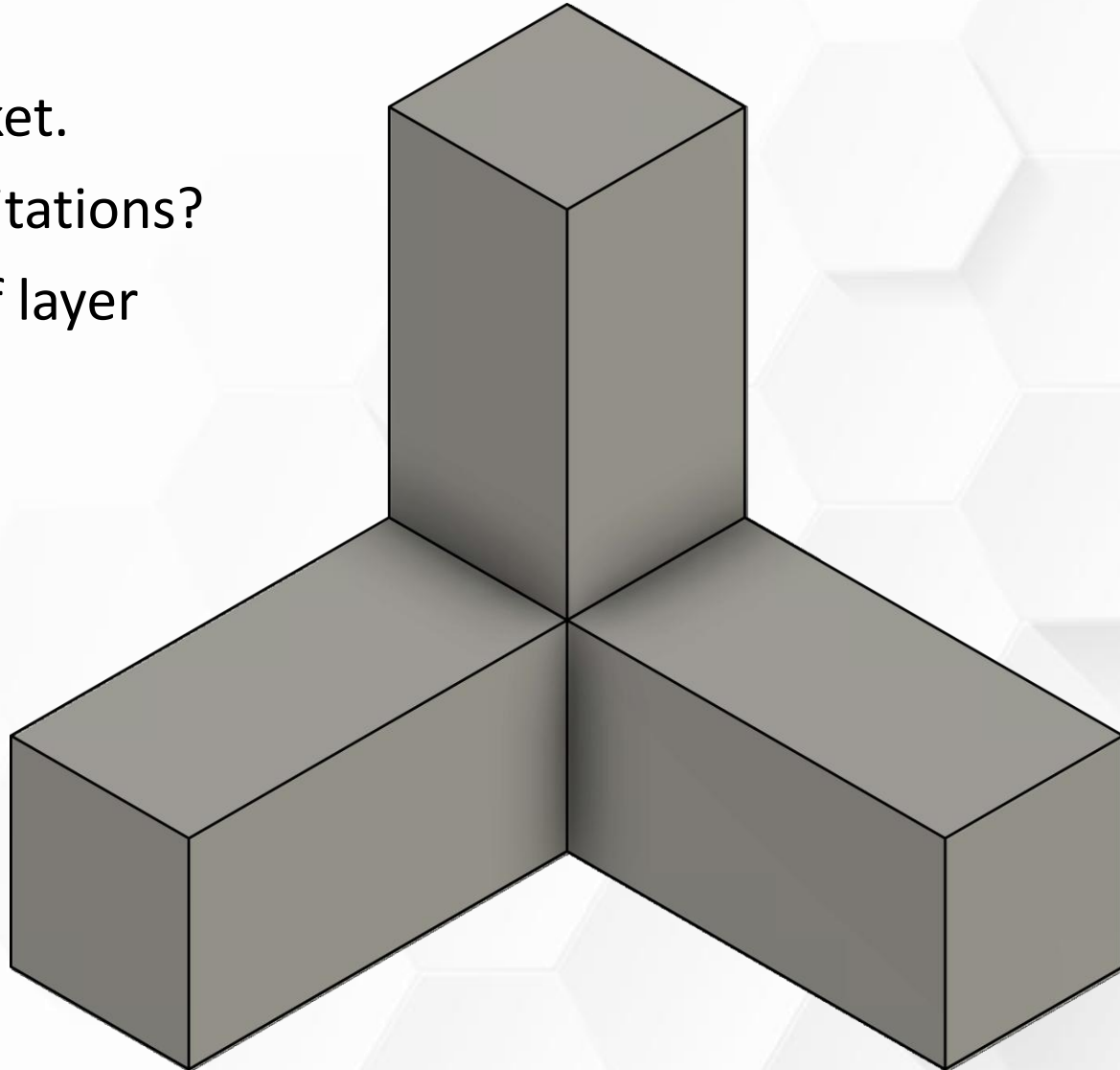
Thermal Expansion

Critical Thinking: Corners

- Rounding corners with fillets is helpful. Why?
- Stresses concentrate in corners.
 - Adding fillets improves strength.
 - Fillets reduce first layer warping.
- Print faster with more accurate movements. Why?
 - Think of a racecar (toolhead) going around a track.
 - Smoothing changes of direction reduces vibrations.

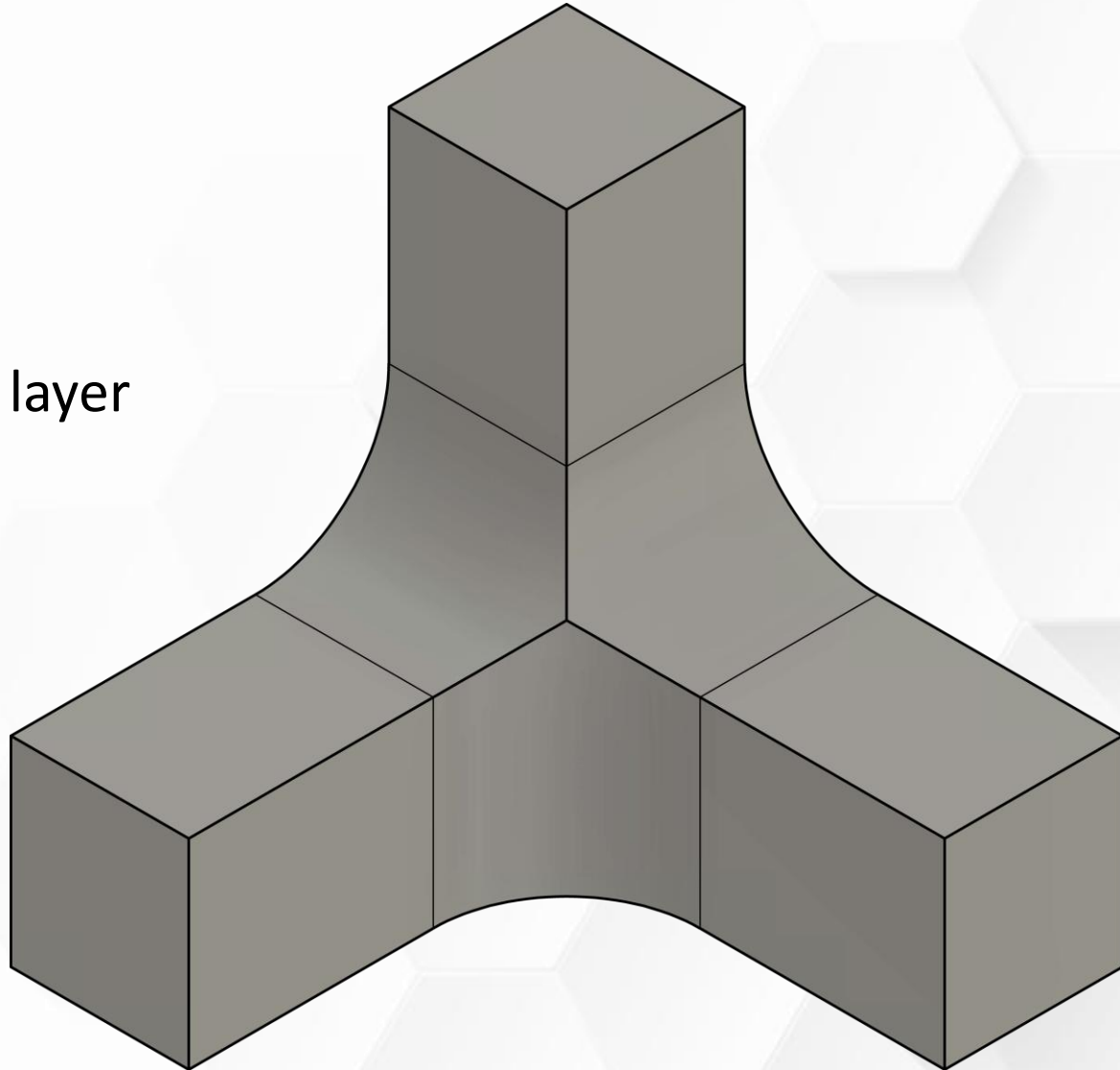
Critical Thinking: Corner Bracket

- Consider this 3-way corner bracket.
- What are the difficulties and limitations?
- Top section is weaker because of layer stacking.
- What can you do?
 - Stronger filament.
 - Stronger slicer settings.
 - What else?



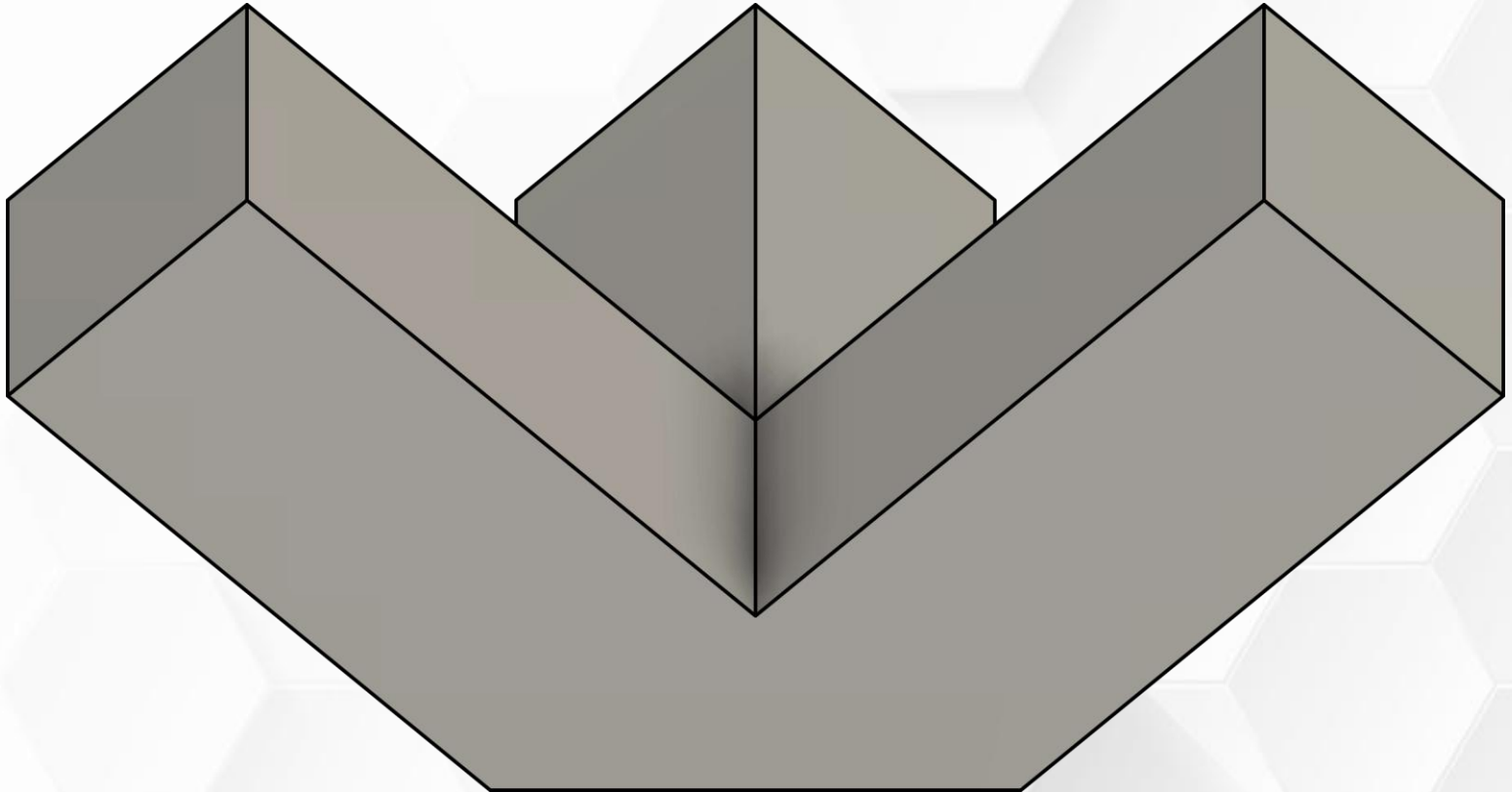
Critical Thinking: Corner Bracket

- Reinforce joints with fillets.
- Reduces stress concentration.
- Shifts the weak point upwards.
- Top section still relies on vertical layer stacking.



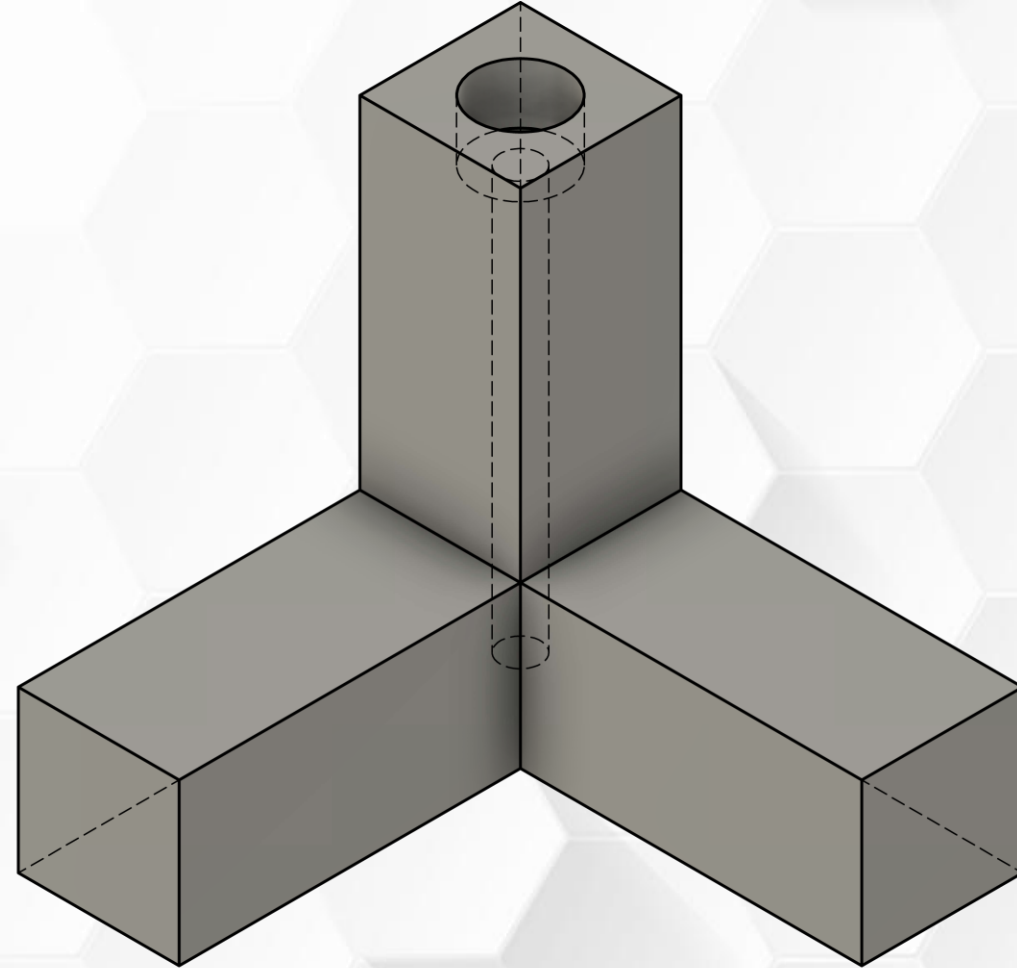
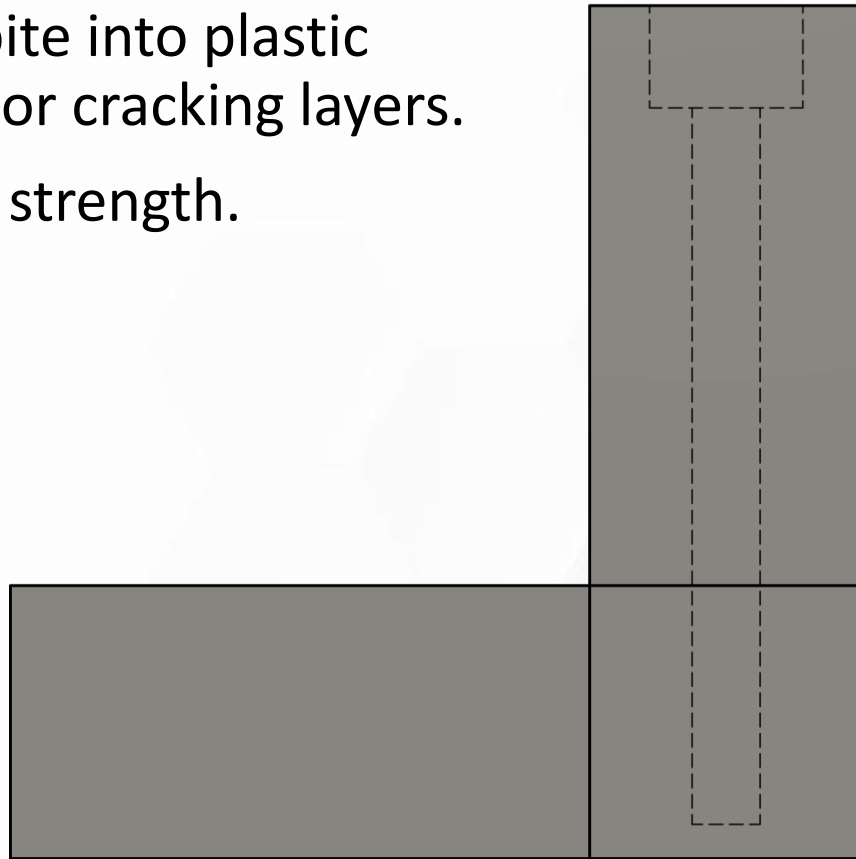
Critical Thinking: Corner Bracket

- Trim the corner to create an angled print orientation.
- This adds overhangs.
- Reduced first layer contact.
- Each section has equal strength.
- Slightly improved strength versus stacking vertical layers.



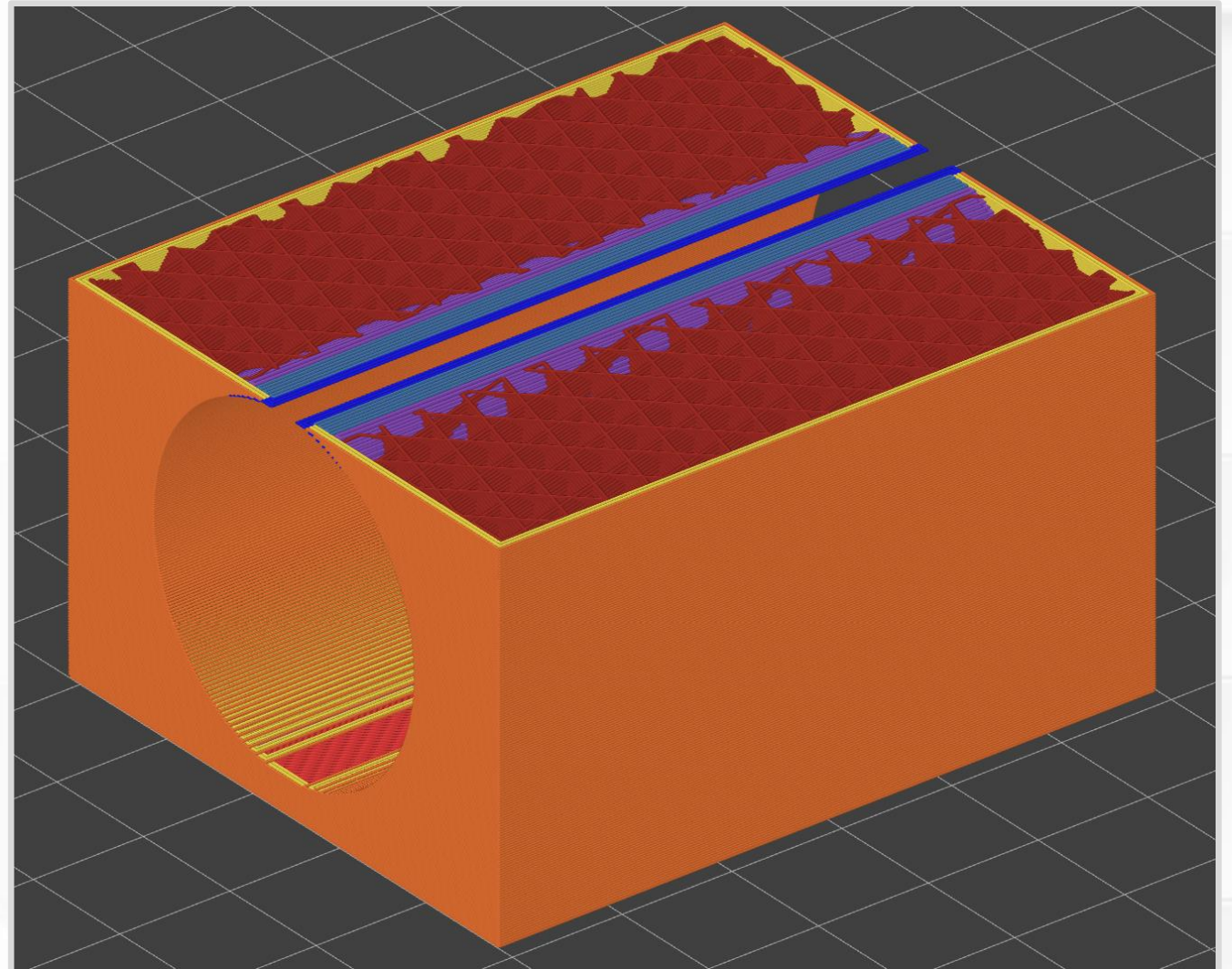
Critical Thinking: Corner Bracket

- Add tight fitting screw after printing.
- Threads should bite into plastic without splitting or cracking layers.
- Large increase in strength.



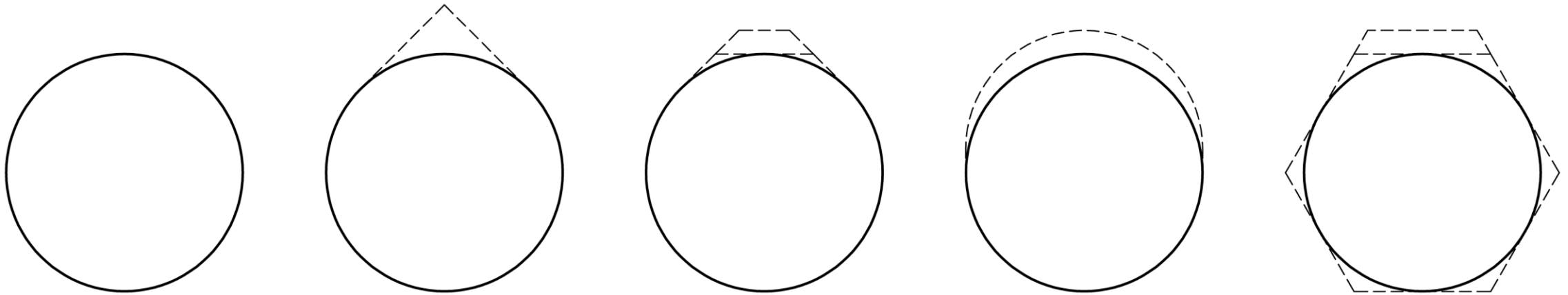
Horizontal Holes

- The top of horizontal holes have extra steep (collapsing) overhangs.
- If these extrusions sag down, the hole will be undersized.



Critical Thinking: Horizontal Holes

- How can you redesign the hole?



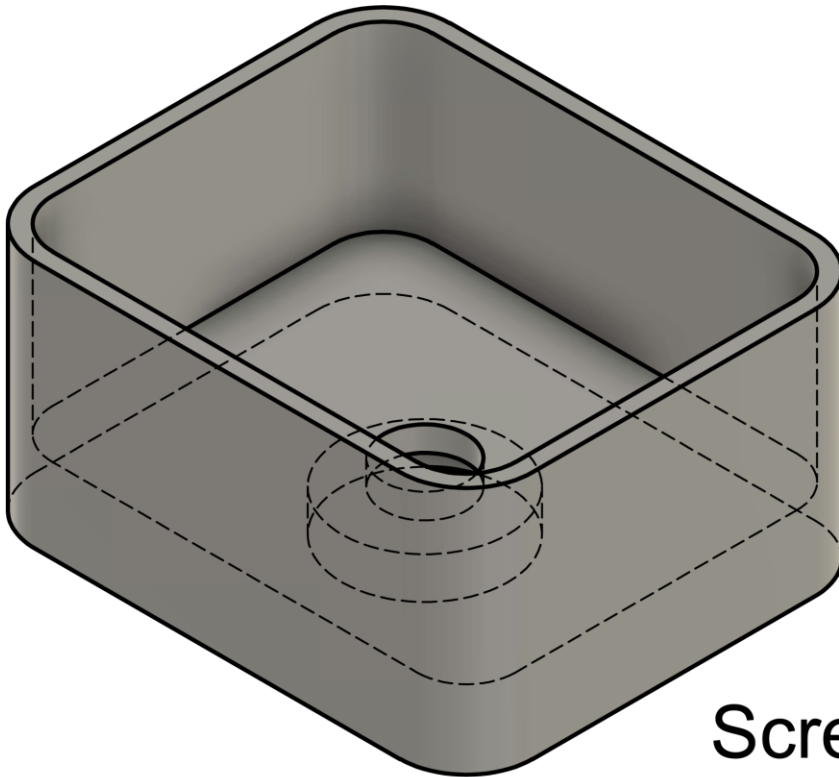
Sequential Bridges

Combine small bridges to create a large bridge.

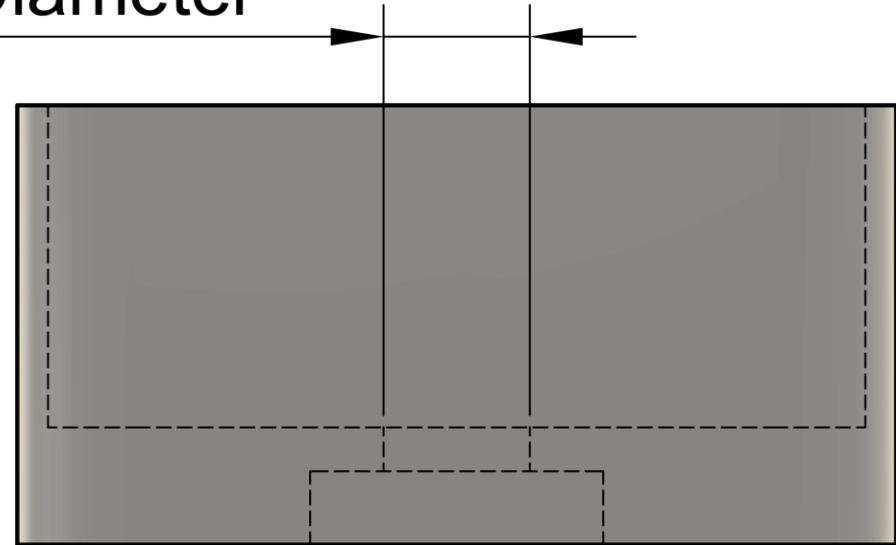


Critical Thinking: Sequential Bridges

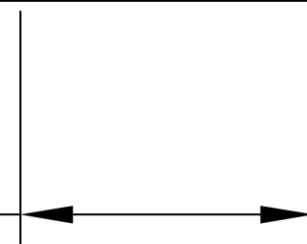
- Storage box with a countersunk screw on the bottom.
- Can you see the hidden problem?



Screw Diameter



Screw Head Diameter



WARNING:

Detected print stability issues:

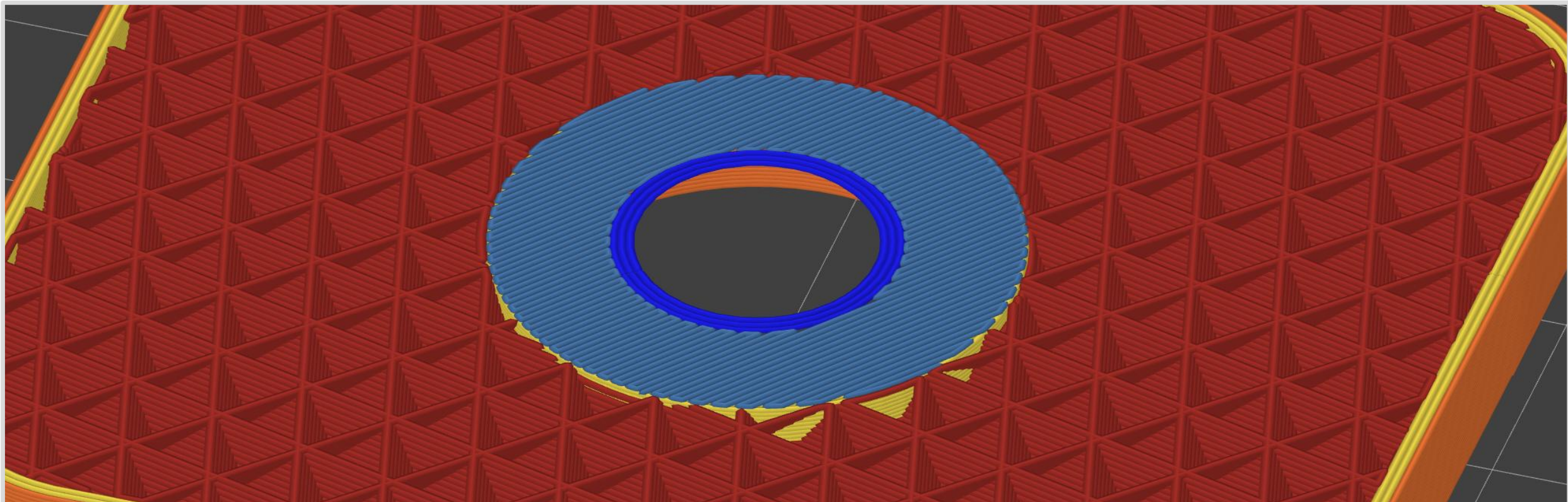


Box Bridge1

Floating bridge anchors, Long bridging extrusions

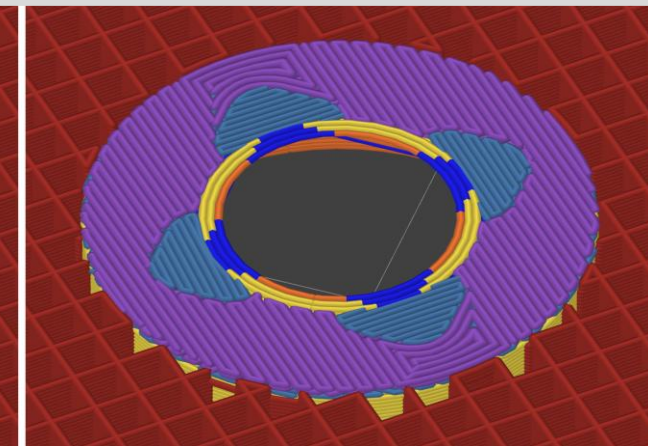
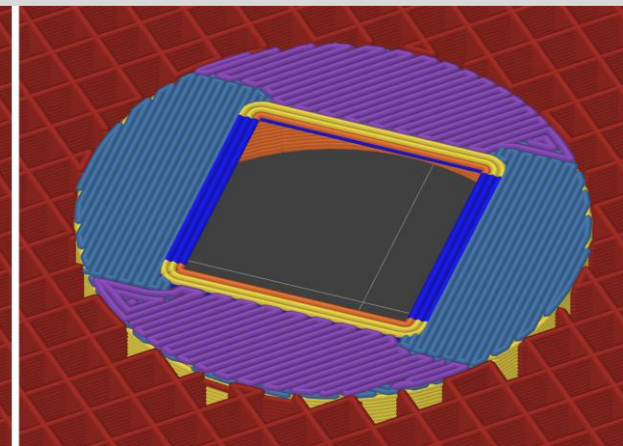
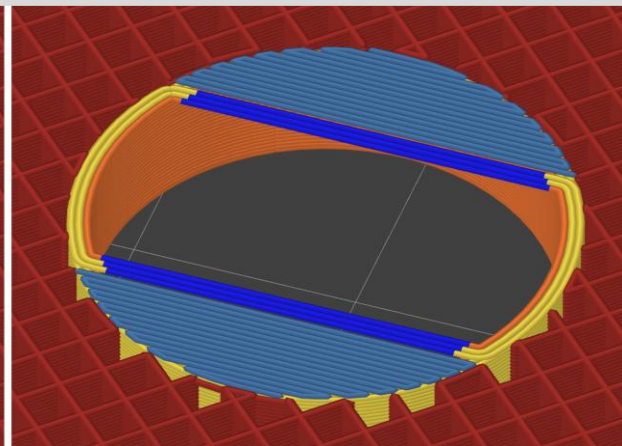
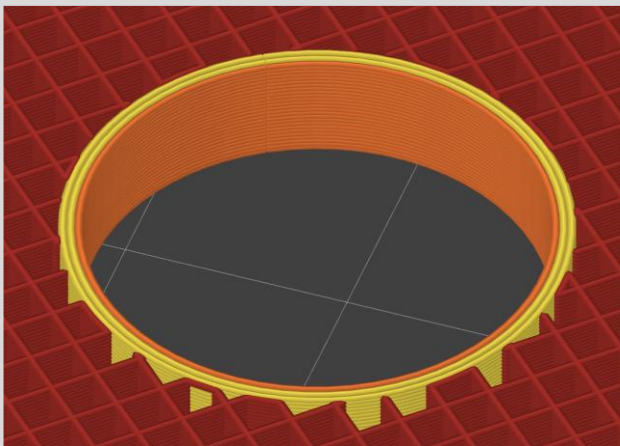
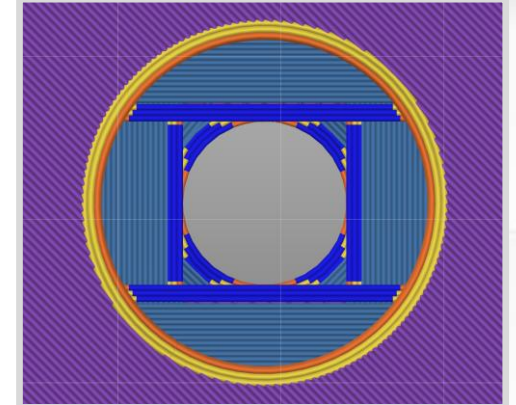
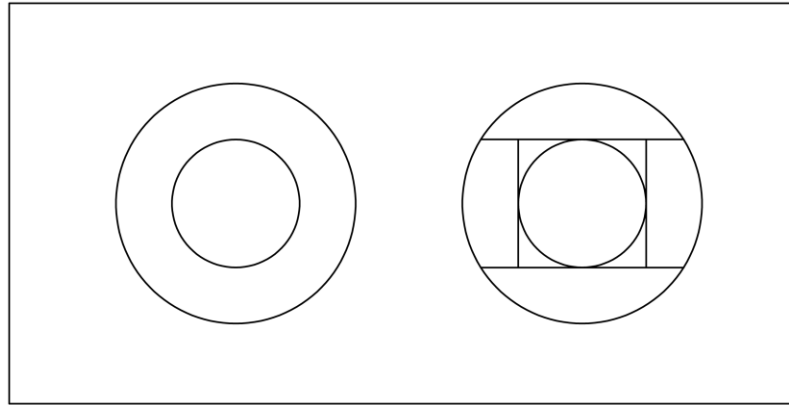


Consider enabling supports.



Critical Thinking: Sequential Bridges

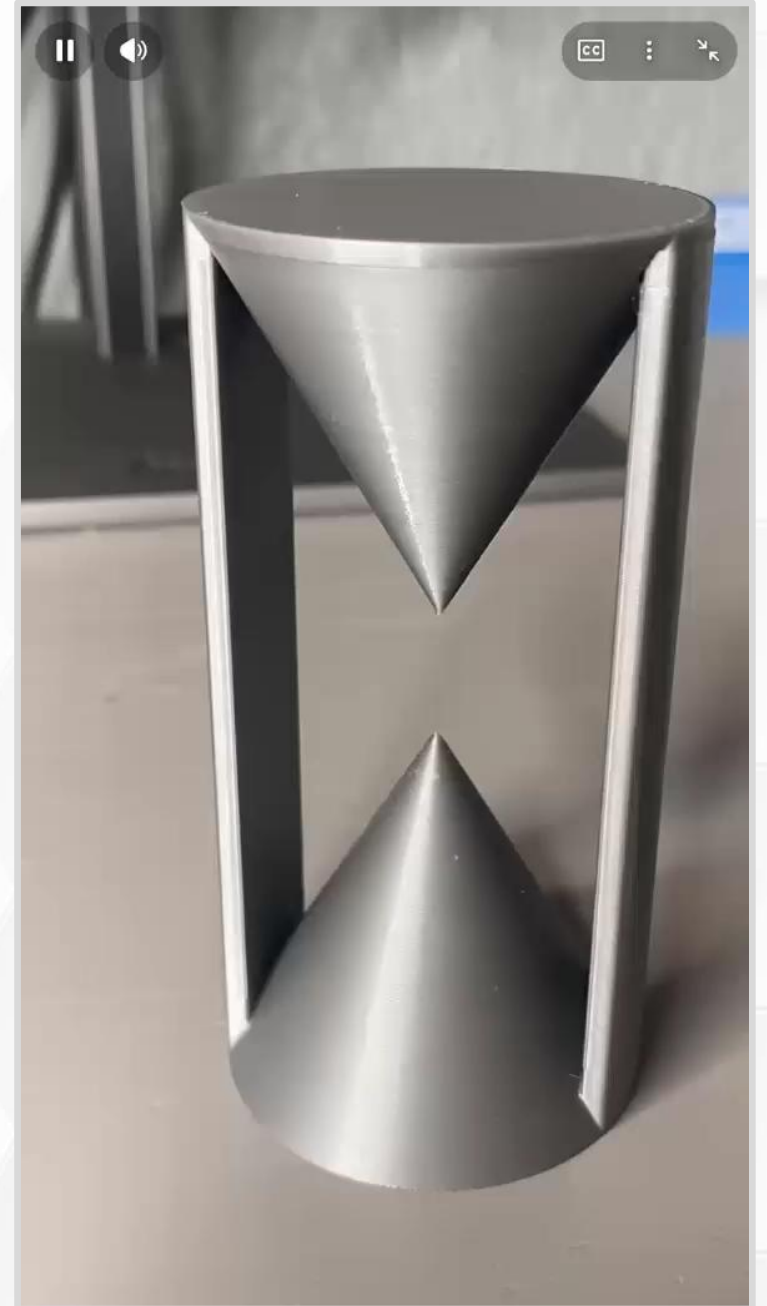
- We can fix this without supports.
- Complete the bridge in multiple steps



Maker Community

- I love the creativity of the maker community.
- This was printed without supports. How?
- Put a printed part inside a printed part.

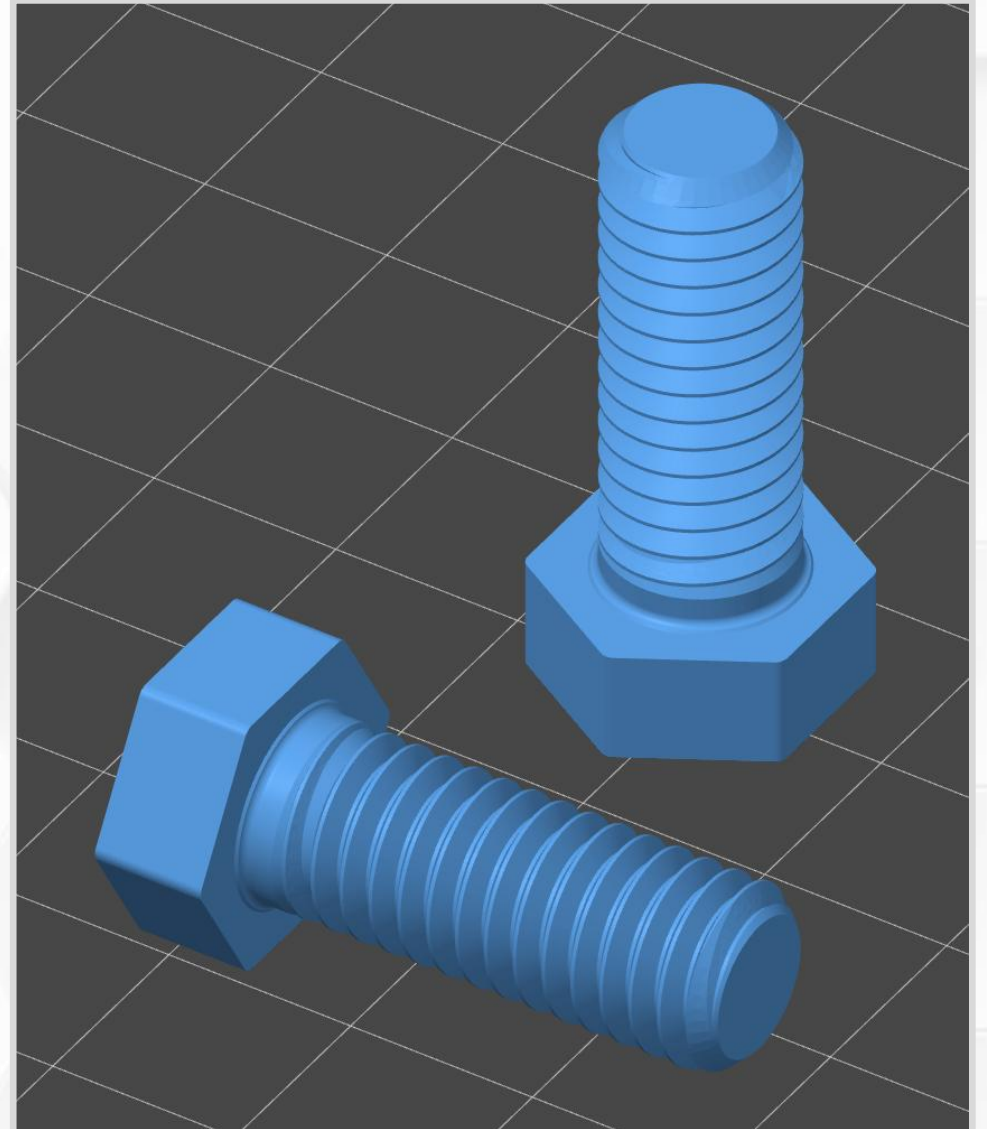
(YouTube short by [christopherhelmke](#))



Critical Thinking: Screws

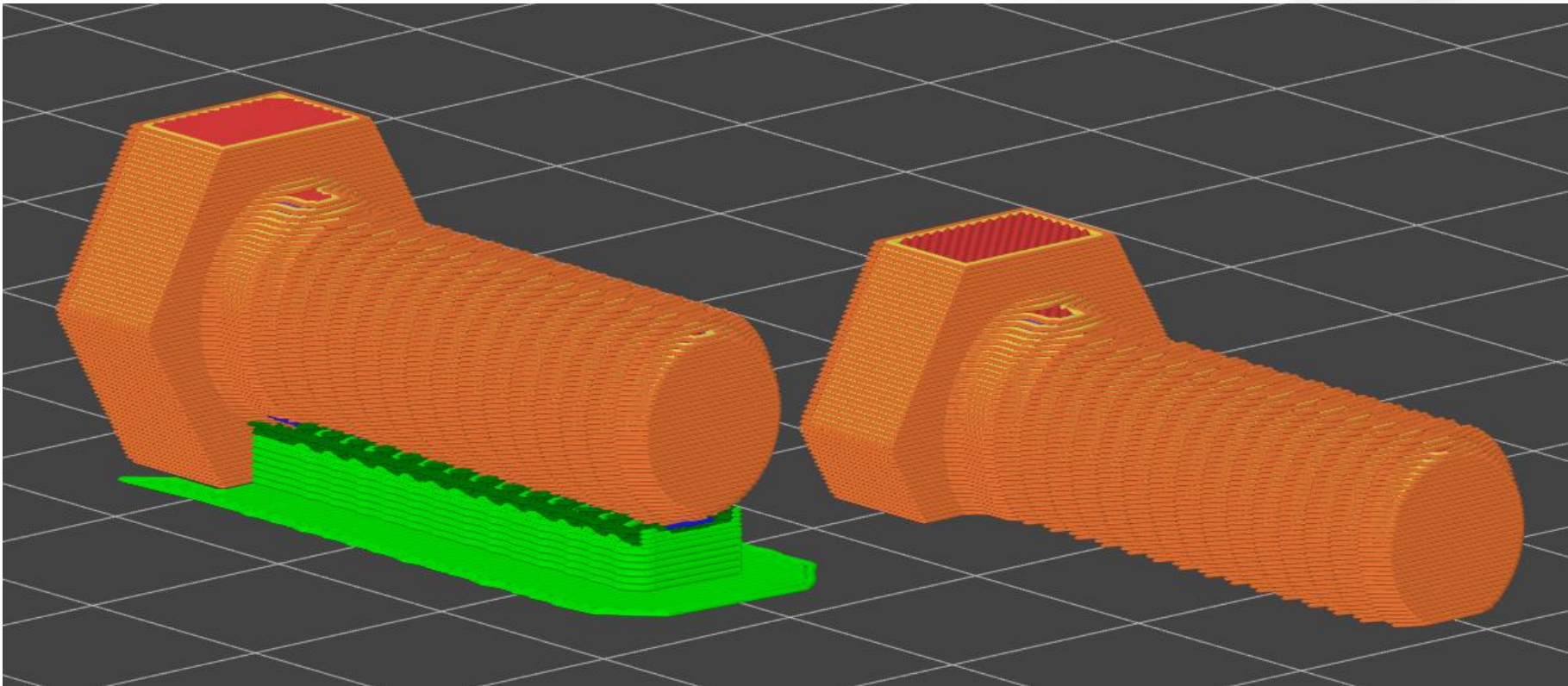
Let's 3D print a plastic screw.

- Compare and contrast print orientations.
- Horizontal.
 - Layer lines maximize strength.
 - Nozzle must be smaller than threads.
 - Bottom threads are floating extrusions.



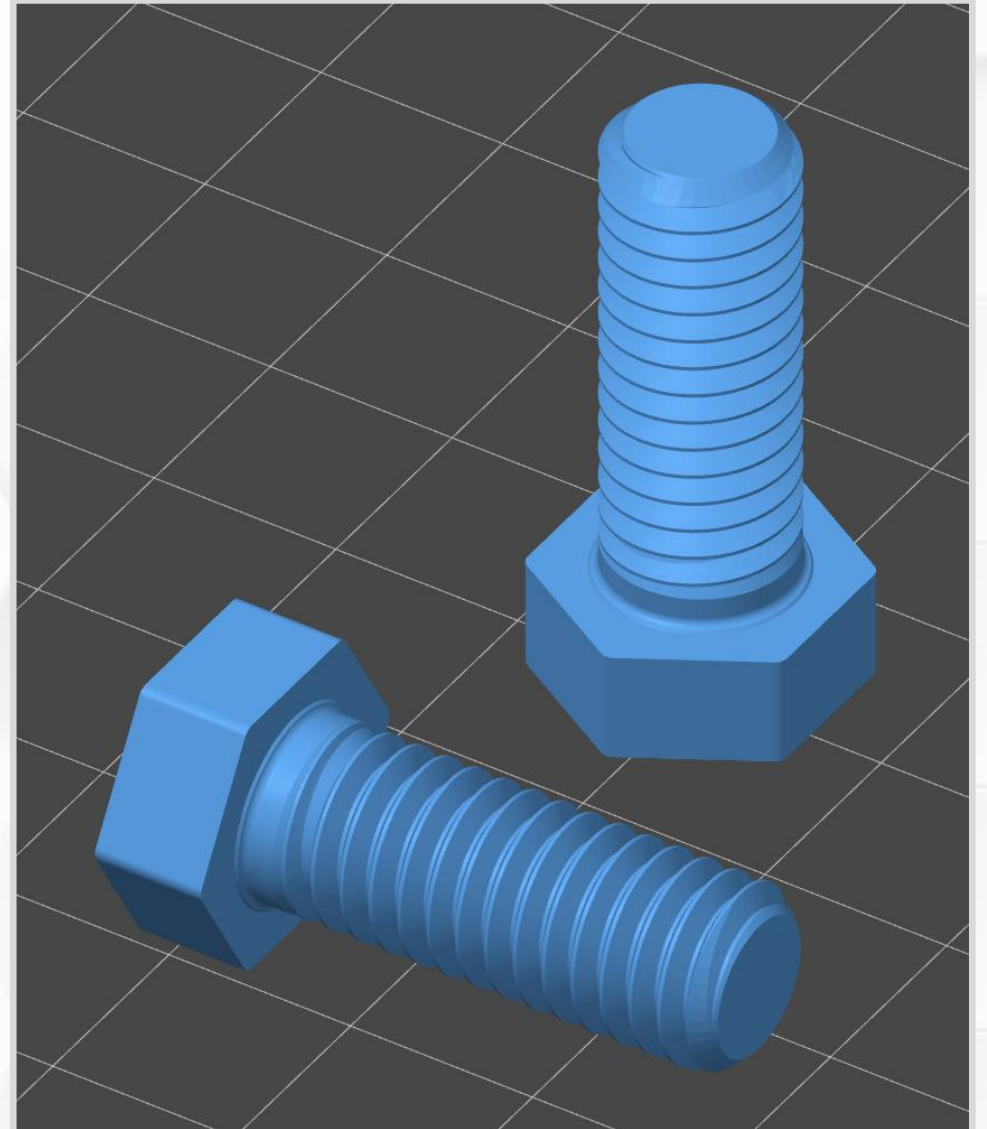
Critical Thinking: Screws

- How can you improve printability?
- Trim the bottom of the bolt so that it sits flat.



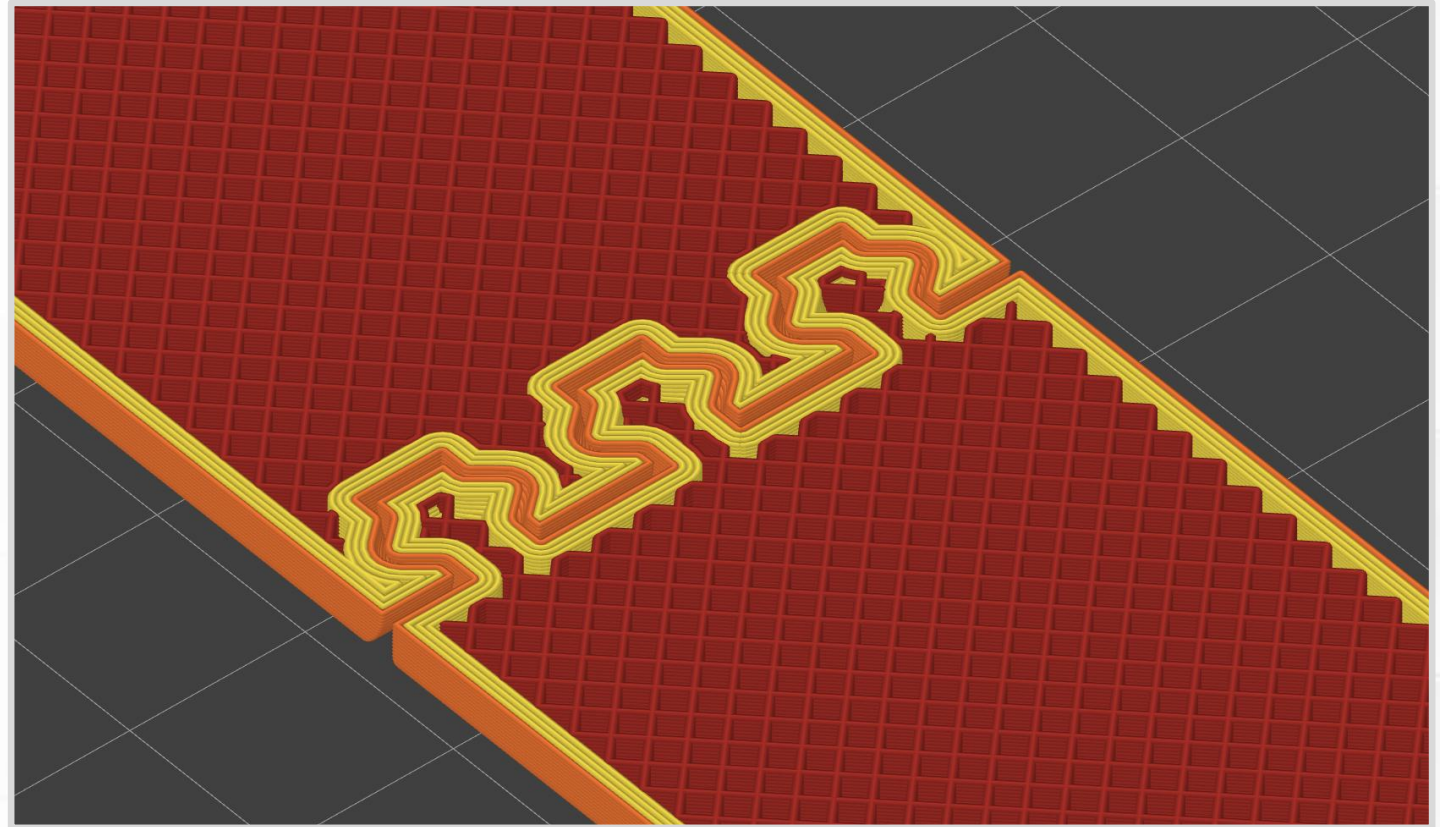
Critical Thinking: Screws

- Vertical
 - Threads limited by layer height.
 - Steep overhangs.
 - Layers lines create a weak part.
 - No supports.
 - No modifications required.
- Threads optimized for 3D printing by printables.com/@FlyingGyroscope.
- Strength testing printed bolts by [My Tech Fun](#). And a [follow-up video](#).



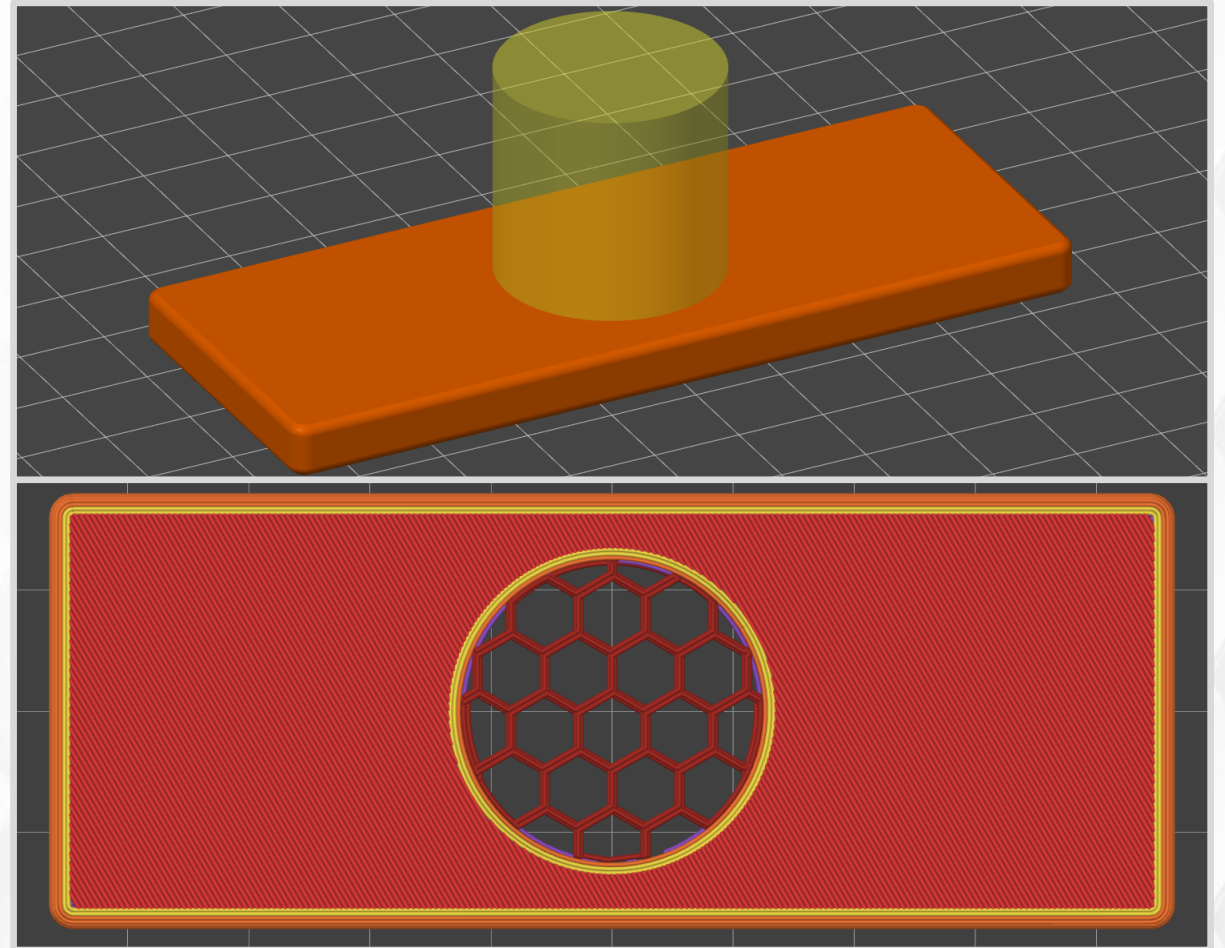
Modeling with a Slicer

- Let's make a print-in-place hinge in PrusaSlicer.
- Flexify template. Quick and easy hinges for beginners.
- Start with a flat object.
- Use negative volume to cut away material.



Modeling with a Slicer

- Let's make a mesh grill using PrusaSlicer.
- Define custom settings for a specific region with a **Modifier**.
- Assign zero top and bottom layers.
- Set the infill pattern.
- Change infill percent to alter the mesh size.



Critical Thinking: Snap Buckle

Let's design a replacement part.

- Should I print a direct copy?
- What concepts and lessons from today can you apply here?
- How can you design a strong buckle?
- Compare and contrast vertical and horizontal print orientation.



(Picture from [HomeDepot.](#))

Critical Thinking: Snap Buckle

- What choices did the designer make?



(Picture from [HomeDepot.](#))



([Snap Buckle by Squinn.](#))

Critical Thinking: Snap Buckle

- Print parts laying down with horizontal print lines.
 - Good strength for the compliant mechanism.
 - Good first layer contact.
- The female socket has been reimagined.
 - Overhangs, not bridges.
 - The middle prong is a dovetail (trapezoid).



(Snap Buckle by Squinn.)